

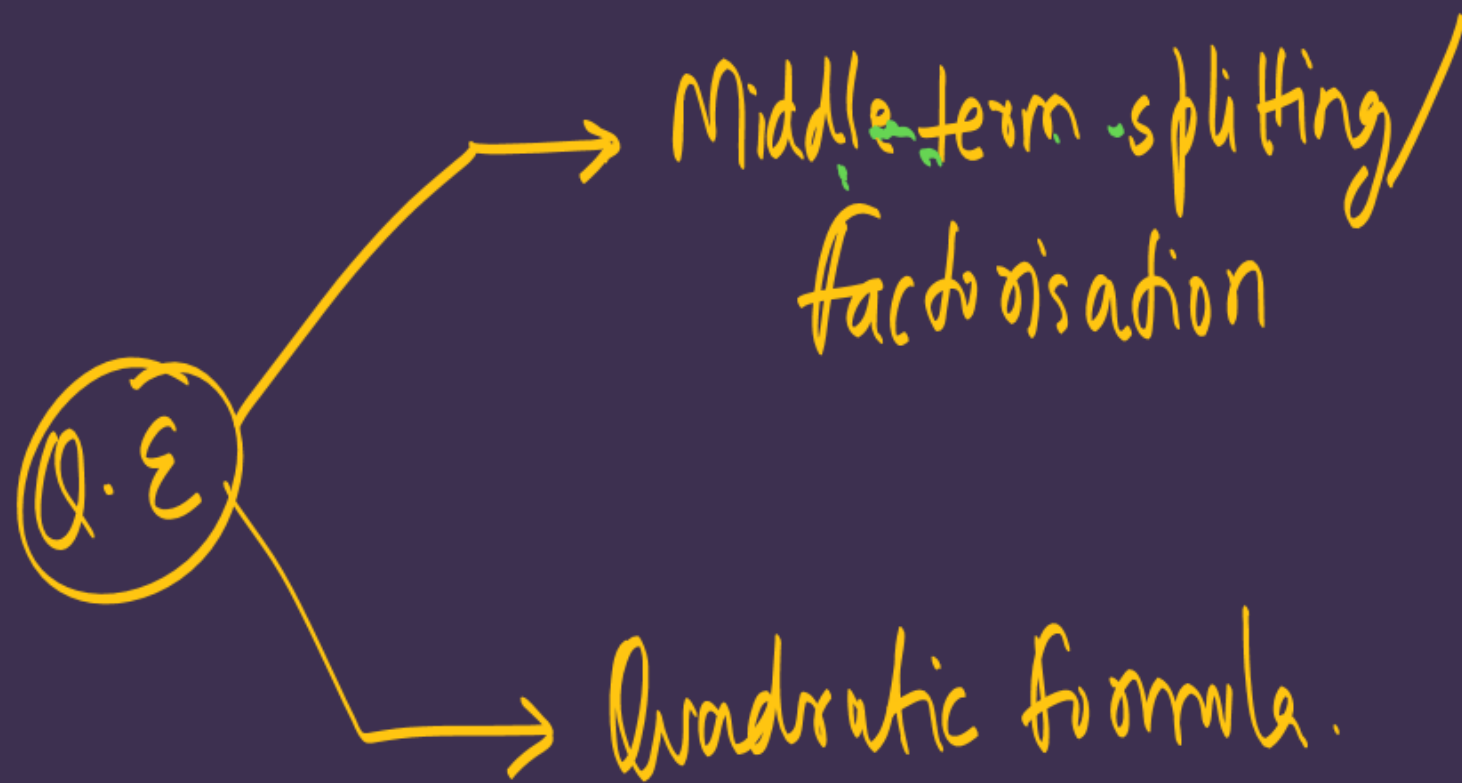
[Aim: 100/100 in Maths]

आरंभ

QUADRATIC EQUATIONS

Lecture #5

ATK² :-



$$x = \frac{-b \pm \sqrt{D}}{2a}$$

Nature of Roots:-

$$D = b^2 - 4ac$$

$$\begin{aligned} D > 0 &\rightarrow \text{Real \& distinct} \\ D = 0 &\rightarrow \text{Real \& Equal} \\ D < 0 &\rightarrow \text{Unreal roots} \end{aligned}$$

K3B → "Real" roots

$$D \geq 0$$

एक Real roots
एक

Solve

#LP: $\frac{m}{n}x^2 + \frac{n}{m} = 1 - 2x$

(CBSE 2017)

$$\Rightarrow \frac{m}{n}x^2 + \frac{n}{m} - 1 + 2x = 0$$

$$\Rightarrow \frac{m^2x^2 + n^2 - 1(nm) + 2x(nm)}{nm} = 0$$

$$\Rightarrow m^2x^2 + n^2 + 2xnm - nm = 0 \times nm$$

$$\Rightarrow (mx)^2 + (n)^2 + 2(mx)(n) - nm = 0$$

$(A)^2 + (B)^2 + 2(A)(B)$
 $\Rightarrow (A+B)^2$

$$\Rightarrow (mx+n)^2 - (\sqrt{nm})^2 = 0$$

$$\Rightarrow (mx+n)^2 - (\sqrt{nm})^2 = 0$$

$A^2 - B^2$
 $(A+B)(A-B)$

$$\Rightarrow (mx+n+\sqrt{nm})(mx+n-\sqrt{nm}) = 0$$

$$\Rightarrow mx+n+\sqrt{nm}=0$$

$$\Rightarrow mx = -n - \sqrt{nm}$$

$$x = \frac{-n - \sqrt{nm}}{m}$$

$$\Rightarrow mx+n-\sqrt{nm}=0$$

$$\Rightarrow mx = -n + \sqrt{nm}$$

$$x = \frac{-n + \sqrt{nm}}{m}$$

#K3B

$$x^2 - 4x + 4 = 0$$

or, $(x-2)^2 = 0$

$$D = b^2 - 4ac$$
$$= (-4)^2 - 4(1)(4)$$

$$\rightarrow 16 - 16$$

$$D = 0$$

↓
genl & equal roots

$$x^2 + 4 - 4x$$

$$\Rightarrow (x)^2 + (2)^2 - 2(2)(x)$$
$$A^2 + B^2 - 2(B)(A)$$

$$\Rightarrow (x-2)^2$$

Perfect Square

$D=0$ होता है

#LP: For what value of k , $\underbrace{(4-k)}_a x^2 + \underbrace{(2k+4)}_b x + \underbrace{(8k+1)}_c = 0$, is a perfect square?

perfect square $\rightarrow D = 0$
 $b^2 - 4ac = 0$

$$\Rightarrow (2k+4)^2 - 4(4-k)(8k+1) = 0$$

$$\Rightarrow (2k)^2 + (4)^2 + 2(2k)(4) - 4[32k + 4 - 8k^2 - k] = 0$$

$$\Rightarrow \underline{4k^2} + \cancel{16} + \underline{16k} - \underline{128k} - \cancel{16} + \underline{32k^2} + \underline{4k} = 0$$

$$\Rightarrow 36k^2 - 108k = 0$$

$$\Rightarrow (18k)(2k - 6) = 0$$

$$18k = 0$$

$$k = 0$$

$$2k - 6 = 0$$

$$2k = 6$$

$$k = 3$$

$$k = 0 \text{ \& } 3$$

#LP: Find the discriminant of the quadratic equation $4x^2 - 5 = 0$ and hence comment on the nature of the roots of the equation. (CBSE 2023)

$$4x^2 - 5 = 0 \Rightarrow 4x^2 + 0x - 5 = 0$$

$$D = b^2 - 4ac$$
$$= (0)^2 - 4(4)(-5)$$

$$D = 80$$

$\rightarrow D > 0 \rightarrow$ real & distinct roots

#LP: If 2 is a root of the quadratic equation $3x^2 + px - 8 = 0$ and the quadratic equation $4x^2 - 2px + k = 0$ has equal roots, find the value of k .

$x = 2 \rightarrow$ put in $3x^2 + px - 8 = 0$

$$3(2)^2 + p(2) - 8 = 0$$

$$\rightarrow 3(4) + 2p - 8 = 0$$

$$\rightarrow 12 + 2p - 8 = 0$$

$$\rightarrow 4 + 2p = 0$$

$$\rightarrow 2p = -4$$

$$\boxed{p = -2}$$

$$4x^2 - 2px + k = 0$$

$$p = -2 \Rightarrow 4x^2 - 2(-2)x + k = 0$$

$$\boxed{4x^2 + 4x + k = 0}$$

$$\text{Equal roots} \Rightarrow D = 0$$

$$b^2 - 4ac = 0$$

$$\rightarrow (4)^2 - 4(4)(k) = 0$$

$$\rightarrow 16 - 16k = 0$$

$$\rightarrow 16 = 16k$$

$$\rightarrow \boxed{k = 1}$$

#LP: Find the smallest value of p for which the quadratic equation
 $x^2 - 2(p+1)x + p^2 = 0$ has real roots. Hence, find the roots of the equation so
obtained. (CBSE 2025)

Real roots $\rightarrow D \geq 0$

$b^2 - 4ac \geq 0$

$$\Rightarrow [-2(p+1)]^2 - 4(1)(p^2) \geq 0$$

$$\Rightarrow (-2)^2 (p+1)^2 - 4p^2 \geq 0$$

$$\Rightarrow 4(p^2 + 1 + 2p) - 4p^2 \geq 0$$

$$\Rightarrow \cancel{4p^2} + 4 + 8p - \cancel{4p^2} \geq 0$$

$$4 + 8p \geq 0$$

$$8p \geq -4$$

$$p \geq -\frac{4}{8}$$

$$p \geq -\frac{1}{2}$$

"greater than" or "equal"

Smallest value
of $p = -\frac{1}{2}$

$$x^2 - 2(p+1)x + p^2 = 0$$

$$p = -\frac{1}{2} \rightarrow x^2 - 2\left(-\frac{1}{2} + 1\right)x + \left(-\frac{1}{2}\right)^2 = 0$$

$$\Rightarrow x^2 - 2\left(\frac{1}{2}\right)x + \frac{1}{4} = 0$$

$$\Rightarrow x^2 - x + \frac{1}{4} = 0$$

$$\Rightarrow \frac{4x^2 - 4x + 1}{4} = 0$$

$$\Rightarrow \boxed{4x^2 - 4x + 1 = 0}$$

$$4x^2 - 4x + 1 = 0$$

$$\Rightarrow \underbrace{4x^2 - 2x}_{\text{}} - 2x + 1 = 0$$

$$\Rightarrow \underbrace{2x(2x-1)}_{\text{}} - \underbrace{1(2x-1)}_{\text{}} = 0$$

$$\Rightarrow (2x-1)[2x-1] = 0$$

$$\Rightarrow \textcircled{\text{MI}} (2x-1)^2 = 0$$

$$2x-1=0$$

$$2x=1$$

$$\textcircled{x = \frac{1}{2}}$$

$$\textcircled{\text{MII}} \rightarrow$$

$$2x-1=0$$

$$2x=1$$

$$\textcircled{x = \frac{1}{2}}$$

$$2x-1=0$$

$$2x=1$$

$$\textcircled{x = \frac{1}{2}}$$

#LP: If the quadratic equations $2x^2 + 4x + (a + 5) = 0$ have equal roots and $(a + 4)x^2 + ax - 3b = 0$ have distinct real roots, then which of the following is true:

(a) $a = -3, b < 3/4$

(b) $a = 3, b > 3/4$ ✗

✓ (c) $a = -3, b > -3/4$

(d) $a = 3, b < 3/4$ ✗

$$2x^2 + 4x + (a + 5) = 0$$

Equal roots

$$D = 0$$

$$b^2 - 4ac = 0$$

$$\Rightarrow (4)^2 - 4(2)(a + 5) = 0$$

$$\Rightarrow 16 - 8(a + 5) = 0$$

$$\Rightarrow 16 - 8a - 40 = 0$$

$$\Rightarrow -24 - 8a = 0$$

$$\Rightarrow -24 = 8a$$

$$\Rightarrow \boxed{a = -3}$$

$$(ii) (a + 4)x^2 + ax - 3b = 0$$

real & distinct

$$D > 0$$

$$B^2 - 4AC > 0$$

$$\Rightarrow (a)^2 - 4(a + 4)(-3b) > 0$$

$$\Rightarrow a^2 + 4(a + 4)(3b) > 0$$

$$\Rightarrow a^2 + 12b(a + 4) > 0$$

$$\Rightarrow a^2 + 12ab + 48b > 0$$

$$\Rightarrow (-3)^2 + 12(-3)b + 48b > 0$$

$$\Rightarrow 9 - 36b + 48b > 0$$

$$\Rightarrow 9 + 12b > 0$$

$$\Rightarrow 12b > -9$$

$$\Rightarrow b > -\frac{9}{12} \Rightarrow \boxed{b > -3/4}$$

The quadratic equation $\underbrace{(1+a^2)}_A x^2 + \underbrace{2abx}_B + \underbrace{(b^2-c^2)}_C = 0$ has only one root. What is the value of $c^2(1+a^2)$?

$$D = 0$$

$$B^2 - 4AC = 0$$

$$\Rightarrow (2ab)^2 - 4(1+a^2)(b^2-c^2) = 0$$

$$\Rightarrow 4a^2b^2 - 4[b^2 - c^2 + a^2b^2 - a^2c^2] = 0$$

$$\Rightarrow \cancel{4a^2b^2} - \cancel{4b^2} + \cancel{4c^2} - \cancel{4a^2b^2} + \underline{4a^2c^2} = 0$$

$$\Rightarrow -4b^2 + 4c^2 + 4a^2c^2 = 0$$

$$\Rightarrow 4(-b^2 + c^2 + a^2c^2) = 0$$

$$\Rightarrow -b^2 + c^2 + a^2c^2 = 0$$

Having two equal roots

$$D = 0$$

$$c^2 + a^2c^2 = b^2$$

$$\underline{c^2(1+a^2) = b^2}$$

$$\text{Ans} \Rightarrow \boxed{b^2}$$

If the roots of the equation $\underbrace{(a^2 + b^2)}_A x^2 - \underbrace{2(ac + bd)}_B x + \underbrace{(c^2 + d^2)}_C = 0$ are equal, prove that $\frac{a}{b} = \frac{c}{d}$

$$\boxed{D=0}$$

$$B^2 - 4AC = 0$$

$$\Rightarrow [-2(ac + bd)]^2 - 4(a^2 + b^2)(c^2 + d^2) = 0$$

$$\Rightarrow 4(ac + bd)^2 - 4[a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2] = 0$$

$$\Rightarrow 4 \{ \cancel{a^2c^2} + \cancel{b^2d^2} + \underline{2abcd} - \cancel{a^2c^2} - \cancel{a^2d^2} - \cancel{b^2c^2} - \cancel{b^2d^2} \} = 0$$

$$\Rightarrow -\underline{a^2d^2} - \underline{b^2c^2} + \underline{2abcd} = 0$$

$$\Rightarrow \left[\underbrace{(ad)^2}_{A^2} + \underbrace{(bc)^2}_{B^2} - 2 \underbrace{(ad)}_A \underbrace{(bc)}_B \right] = 0$$

$$(A - B)^2 = 0$$

$$(ad - bc)^2 = 0$$

$$ad - bc = 0$$

$$ad = bc$$

$$\frac{a}{b} = \frac{c}{d}$$

Hp

THANK YOU COODIESS !!

